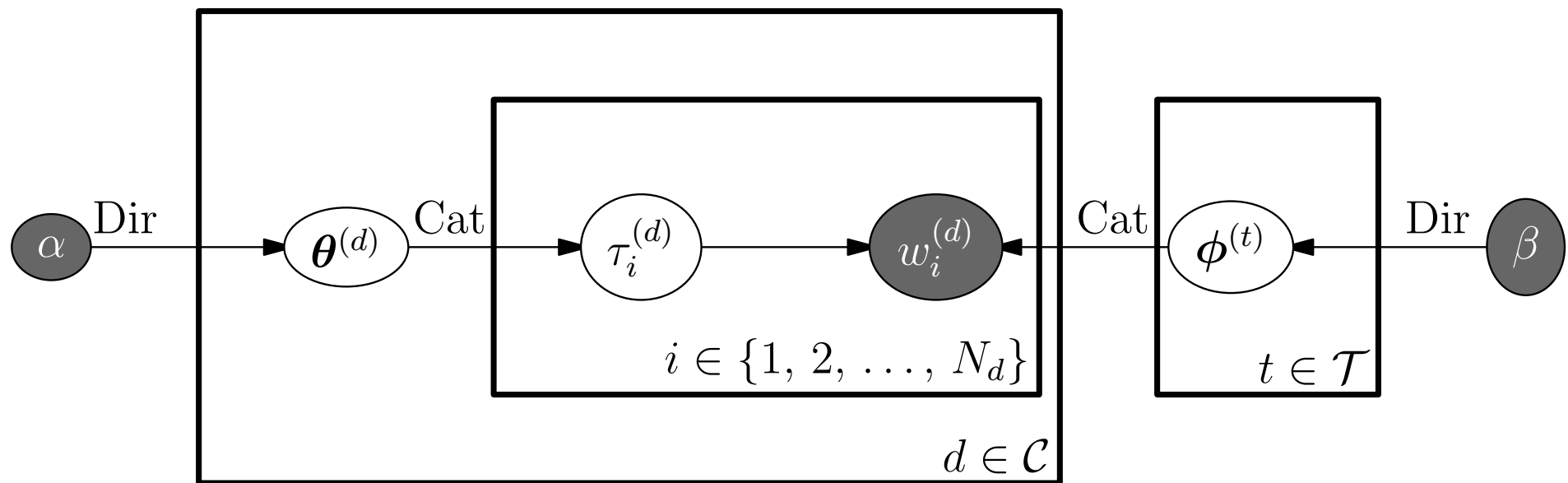


# Advanced Machine Learning

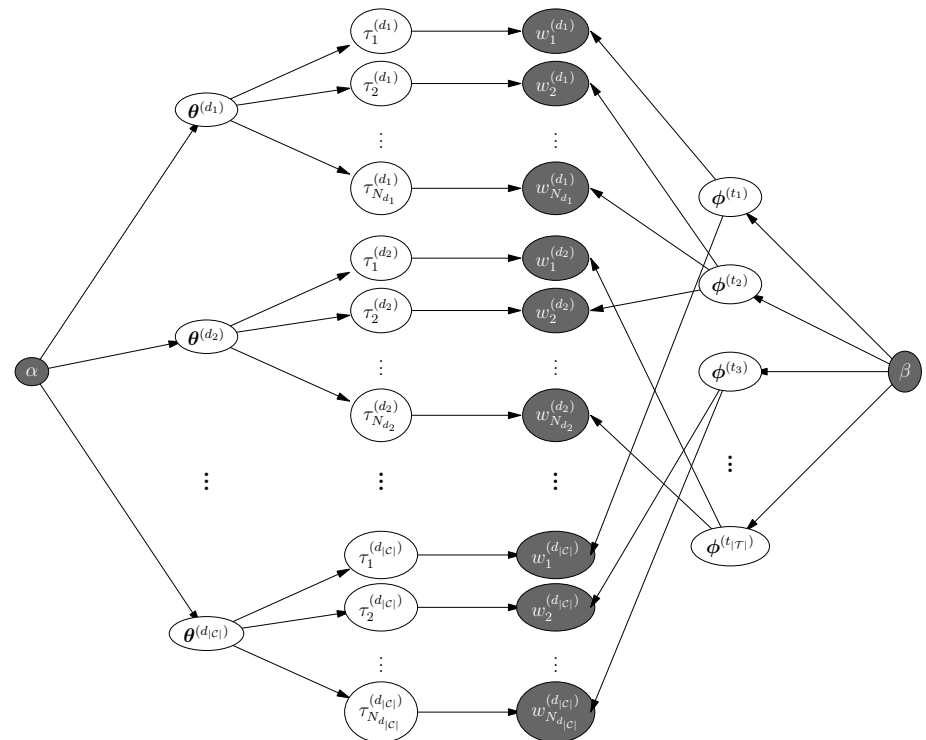
## Graphical Models



*Conditional Independence, Graphical models, LDA*

# Outline

1. Graphical Models
2. Cakes!
3. Latent Dirichlet Allocation



# Graphical Models

- If we want to build large probabilistic inference systems

- ★ AI Doctor

- ★ Fault diagnostic system for a computer

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- In building a probabilistic model we want to know which random variables depend on each other directly and which don't
- Variables that don't will typically still be correlated
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  - ★  $X$  could affect  $Y$
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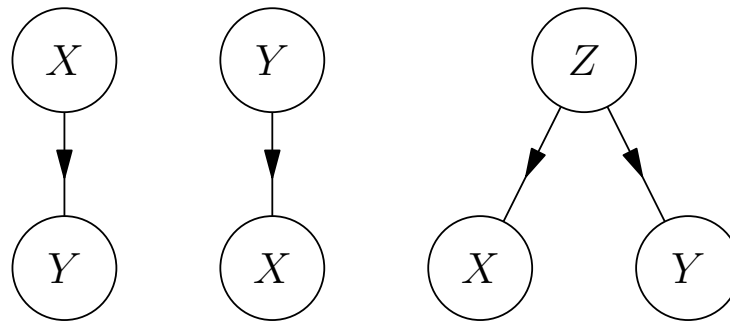
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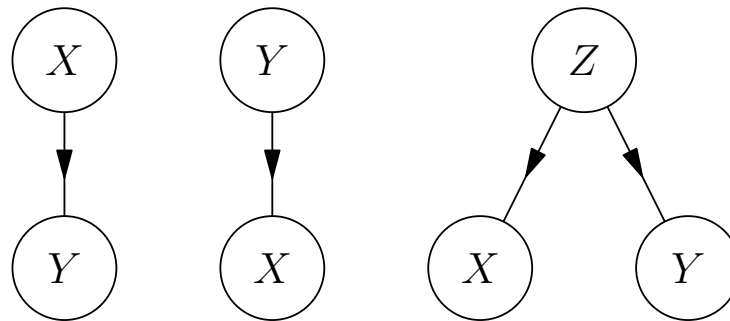
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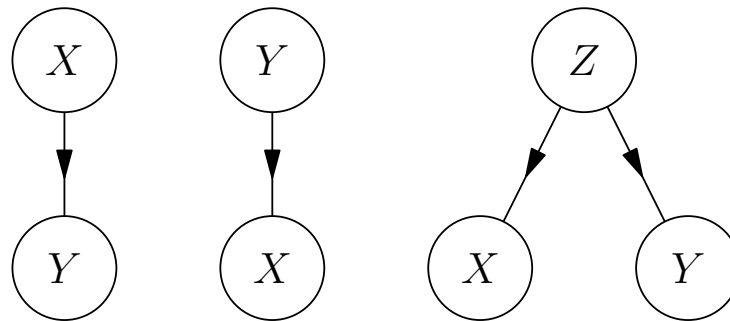
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- Two random variables are statistically independent if

$$\mathbb{P}(X,Y) = \mathbb{P}(X)\mathbb{P}(Y)$$

- Equally this implies  $\mathbb{P}(X|Y) = \mathbb{P}(X)$  and  $\mathbb{P}(Y|X) = \mathbb{P}(Y)$
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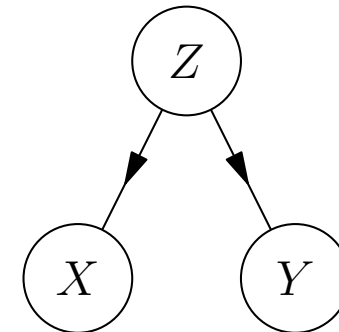
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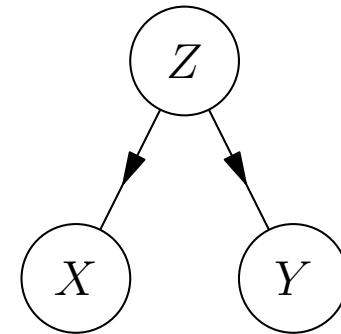
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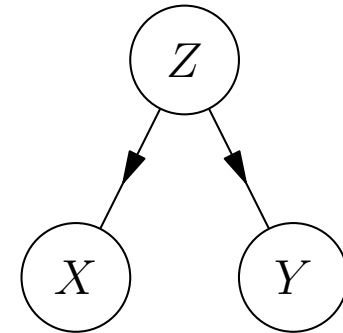
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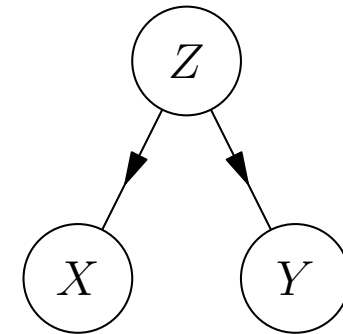
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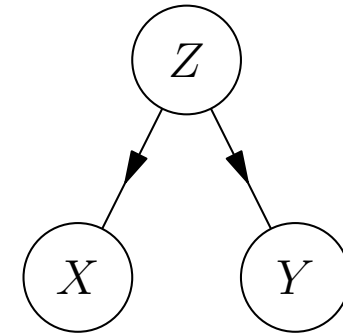
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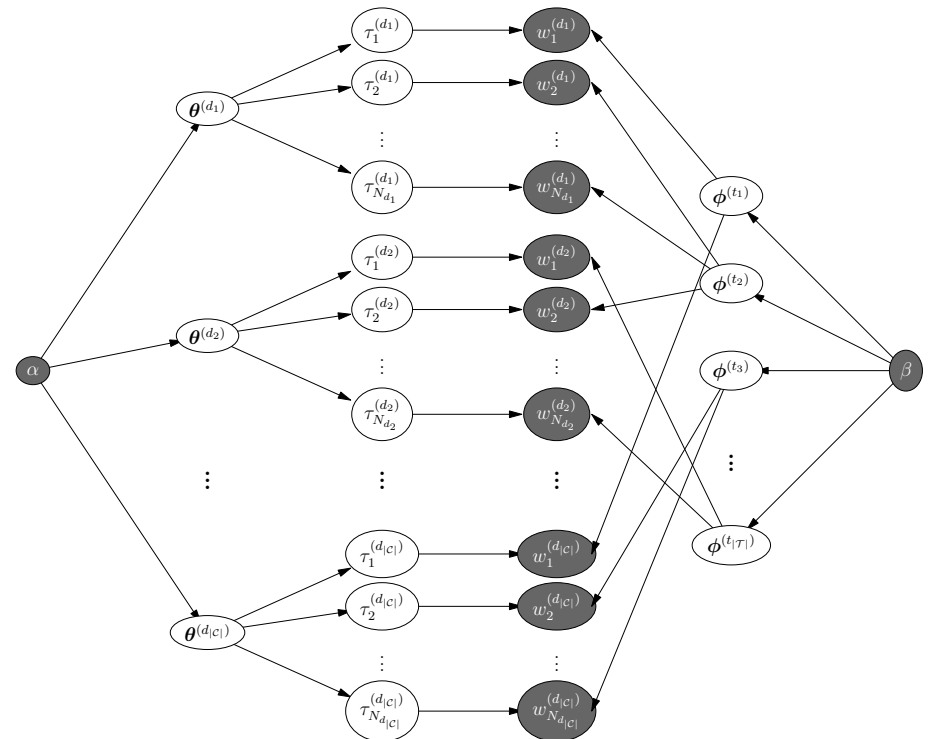


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2. **Cakes!**
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# The Cake Scenario

- Abi and Ben both bake cakes and bring them into the coffee room
- Abi will bring in cakes 20% of the time:  $\mathbb{P}(A = 1) = 0.2$
- Ben will bring in cakes 10% of the time:  $\mathbb{P}(B = 1) = 0.1$
- 90% of the time if either Abi or Ben have put cakes in the coffee room there is some left when I enter  
$$\mathbb{P}(C = 1|A = 1, B = 0) = \mathbb{P}(C = 1|A = 0, B = 1) = 0.9$$
- If they both make cake then there is always cake left  
$$\mathbb{P}(C = 1|A = 1, B = 1) = 1$$
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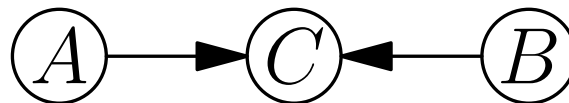
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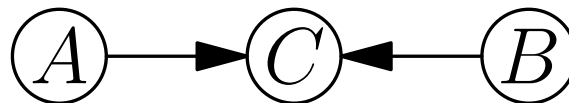


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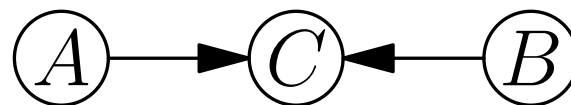
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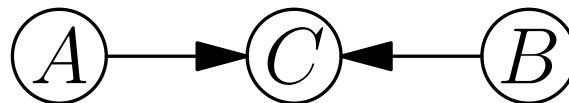
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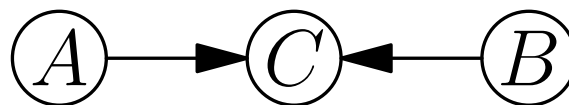
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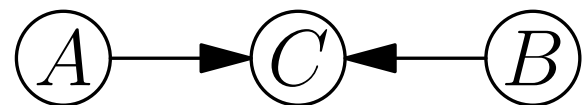
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- By using the joint probability and summing over all unknown quantities, we can compute expectations of anything we are interested in
- These sums are often sped up using knowledge of conditional independence
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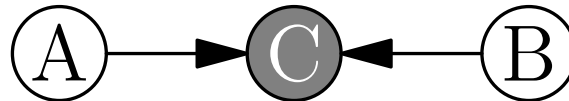
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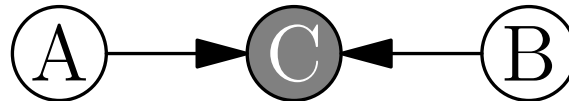
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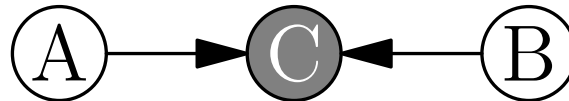
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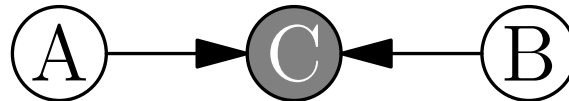
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# Who Made Those Cakes?

- If we observe there are cakes

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- A straightforward if tedious calculation shows

$$\mathbb{P}(A = 1 | C = 1) = 0.628, \quad \mathbb{P}(B = 1 | C = 1) = 0.317$$

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# Elaborate Cakes

- We can elaborate on our cake model
- We suppose that Dave likes cakes so if there is a cake in the coffee room there is a 80% chance that I will see him eating a cake:  $\mathbb{P}(D = 1|C = 1) = 0.8$
- Even if there are no cakes in the coffee room there is a 10% chance that Dave has bought his own cake:  
 $\mathbb{P}(D = 1|C = 0) = 0.1$
- Eli also likes cakes: there is a 60% chance that I will see her eating cakes if there are cakes in the coffee room:  
 $\mathbb{P}(E = 1|C = 1) = 0.6$
- But she never buys herself cakes  $\mathbb{P}(E = 1|C = 0) = 0$

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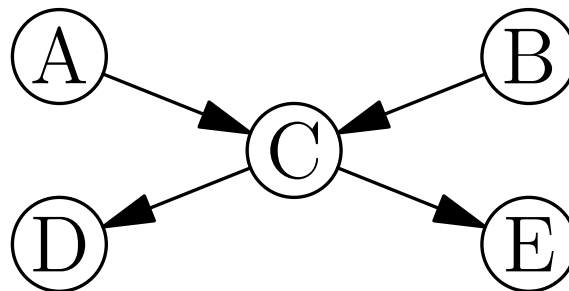
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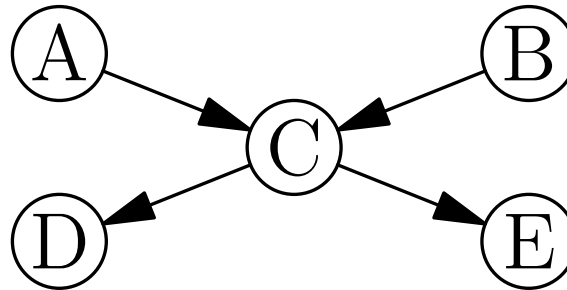
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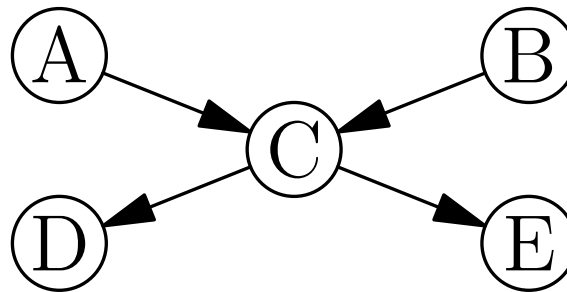
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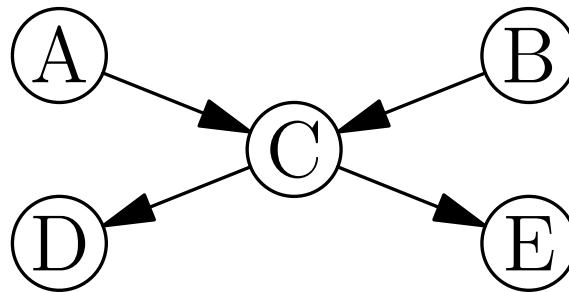
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# Dependencies

- If we don't observe cakes then the probability of Dave and Eli eating cake are not independent

$$\mathbb{P}(D = 1) = 0.303, \quad \mathbb{P}(E = 1) = 0.174$$

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- This changes if we know there are cakes in the coffee room

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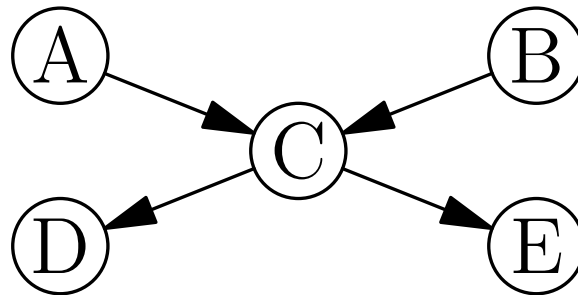
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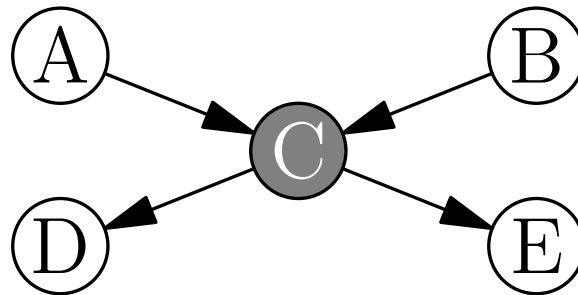


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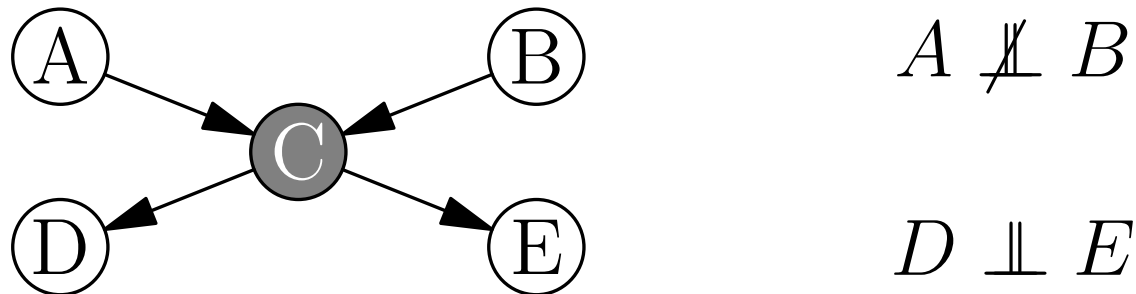


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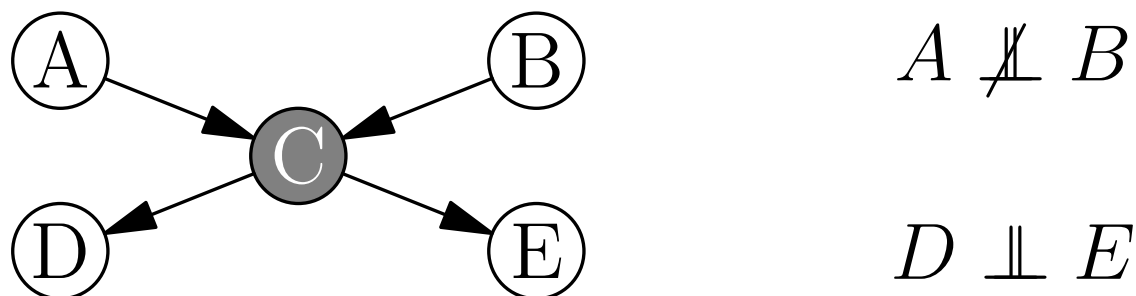
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- There are rules to deduce the conditional independence from a graphical model given which variables have been observed—but these are details that you can look up if needed

# Graphical Model Frameworks

- There are sophisticated frameworks for computing probabilities in Bayesian Belief Networks efficiently
- If our graph is a tree then we can evaluate probabilities efficiently
- When there are loops (so that a random variable both influences and is influenced by another random variables) then exact evaluation of expectations requires exhaustive summing over variables
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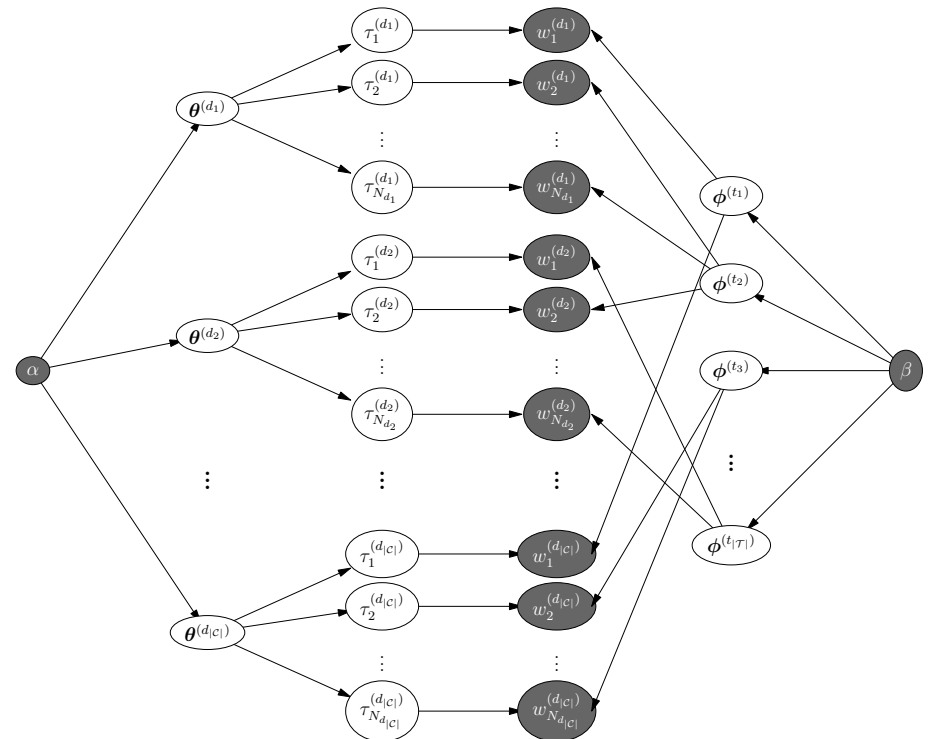
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# Outline

1. Graphical Models
2. Cakes!
3. **Latent Dirichlet Allocation**



# Model for Documents

- We consider a model for the words in a set of documents (we ignore word order)
- We consider a corpus  $\mathcal{C} = \{d_i | i = 1, 2, \dots, |\mathcal{C}|\}$
- With documents consisting of words

$$d = \left( w_1^{(d)}, w_2^{(d)}, \dots, w_{N_d}^{(d)} \right)$$

- We assume that there is a set of topics  $\mathcal{T} = \{t_1, t_2, \dots, t_{|\mathcal{T}|}\}$
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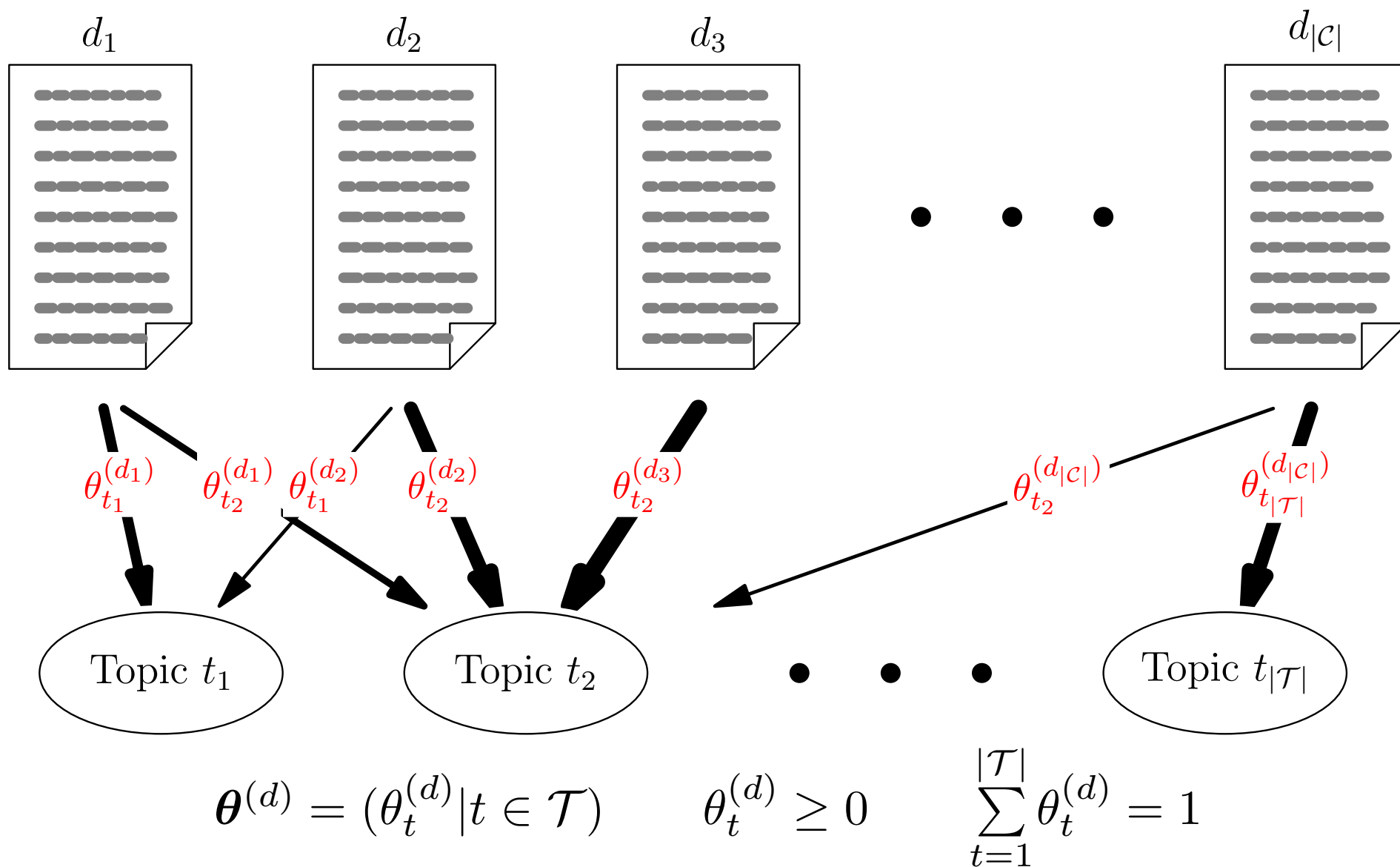
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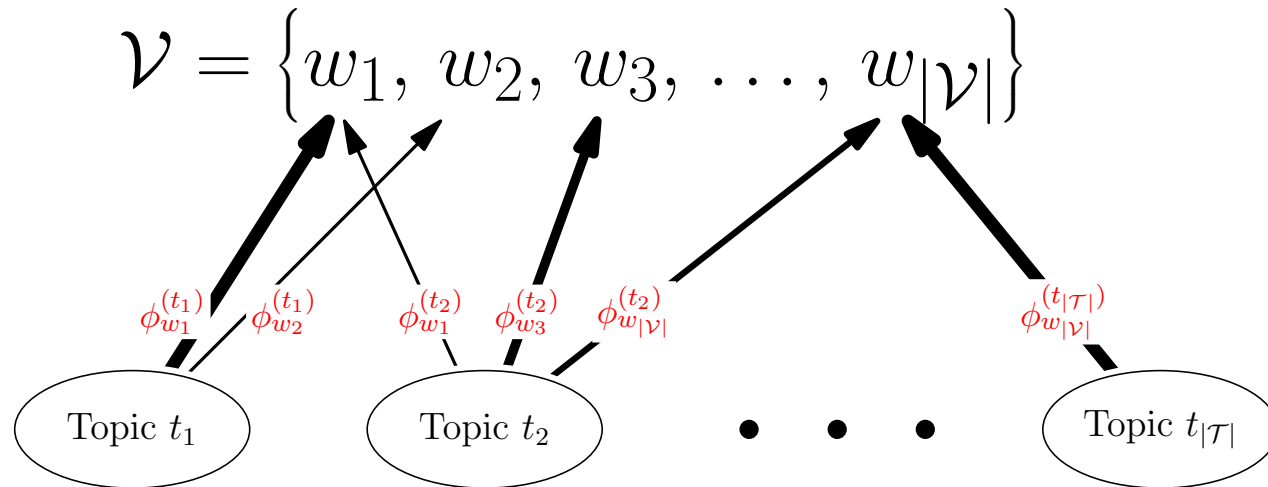
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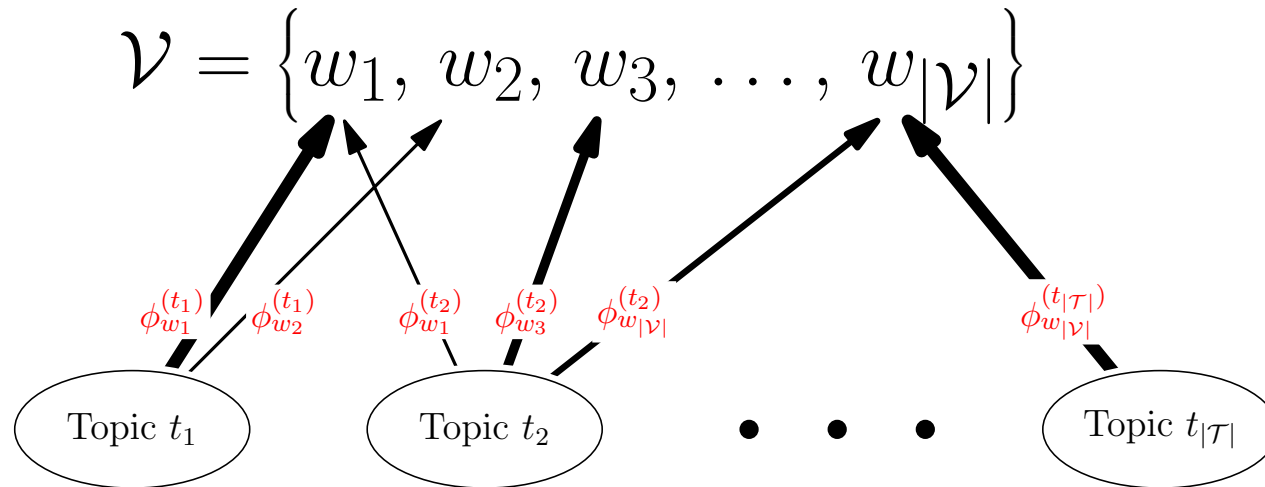
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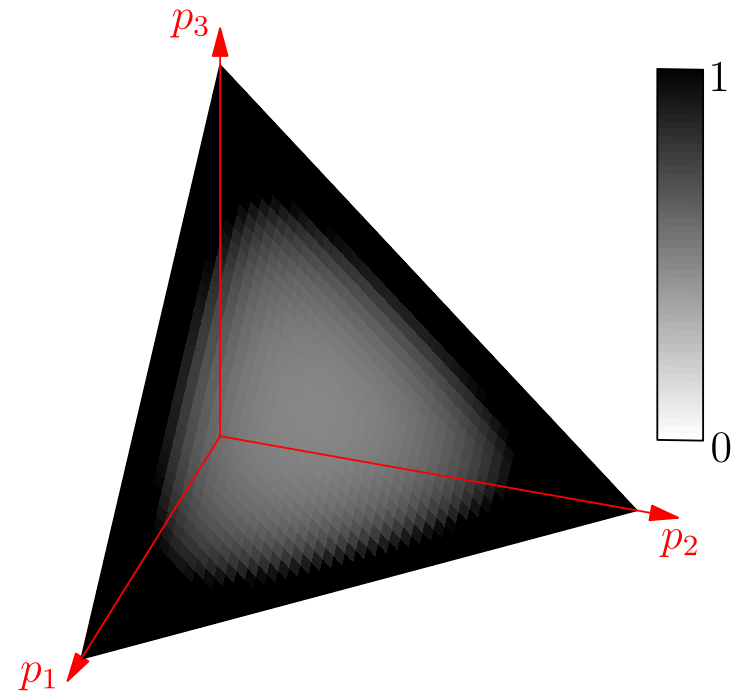
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# Dirichlet Allocation

- Most documents are predominantly about a few topics and most topics have a small number of words associated to them
- We can generate sparse vectors  $\theta^{(d)}$  and  $\phi^{(t)}$  from a Dirichlet distribution with small parameters  $\alpha$

$$\text{Dir}(\mathbf{p}|\boldsymbol{\alpha}) = \Gamma\left(\sum_i \alpha_i\right) \prod_{i=1}^n \frac{p_i^{\alpha_i-1}}{\Gamma(\alpha_i)}$$

- $\sum_i p_i = 1$



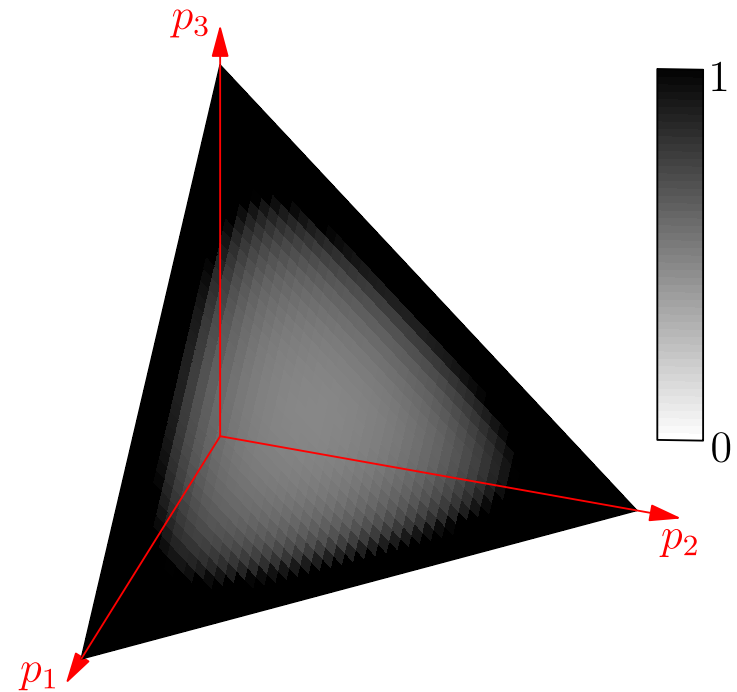
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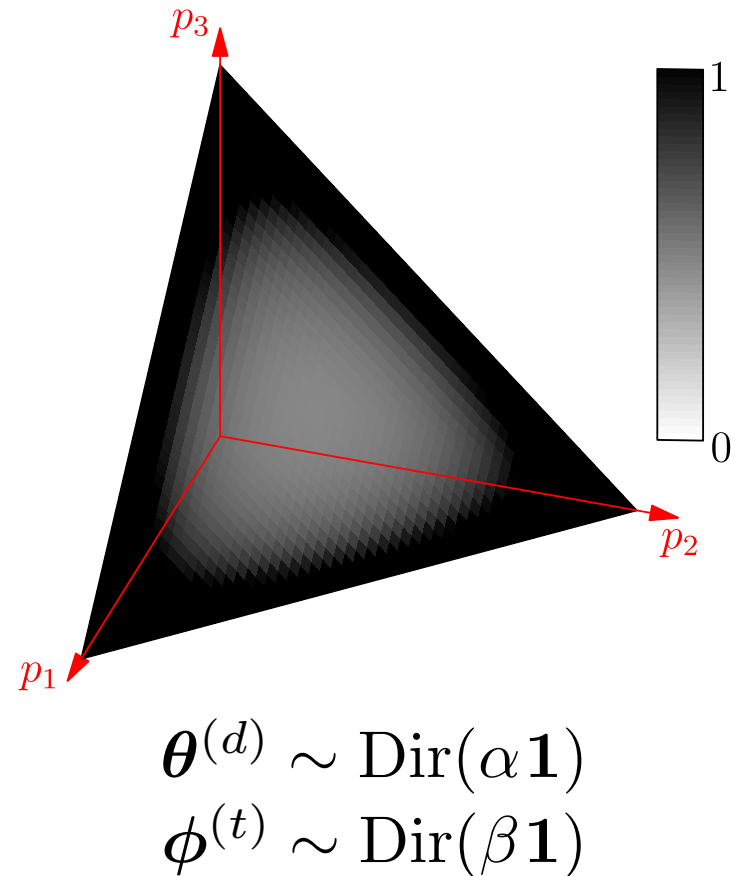
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- Where  $\text{Cat}(i|\mathbf{p}) = p_i$  is the categorical distribution (we choose one of a number of options)
- This model is known as **Latent Dirichlet Allocation**

# Generating Document

- To generate a document we choose a topic for each word and a word for each topic

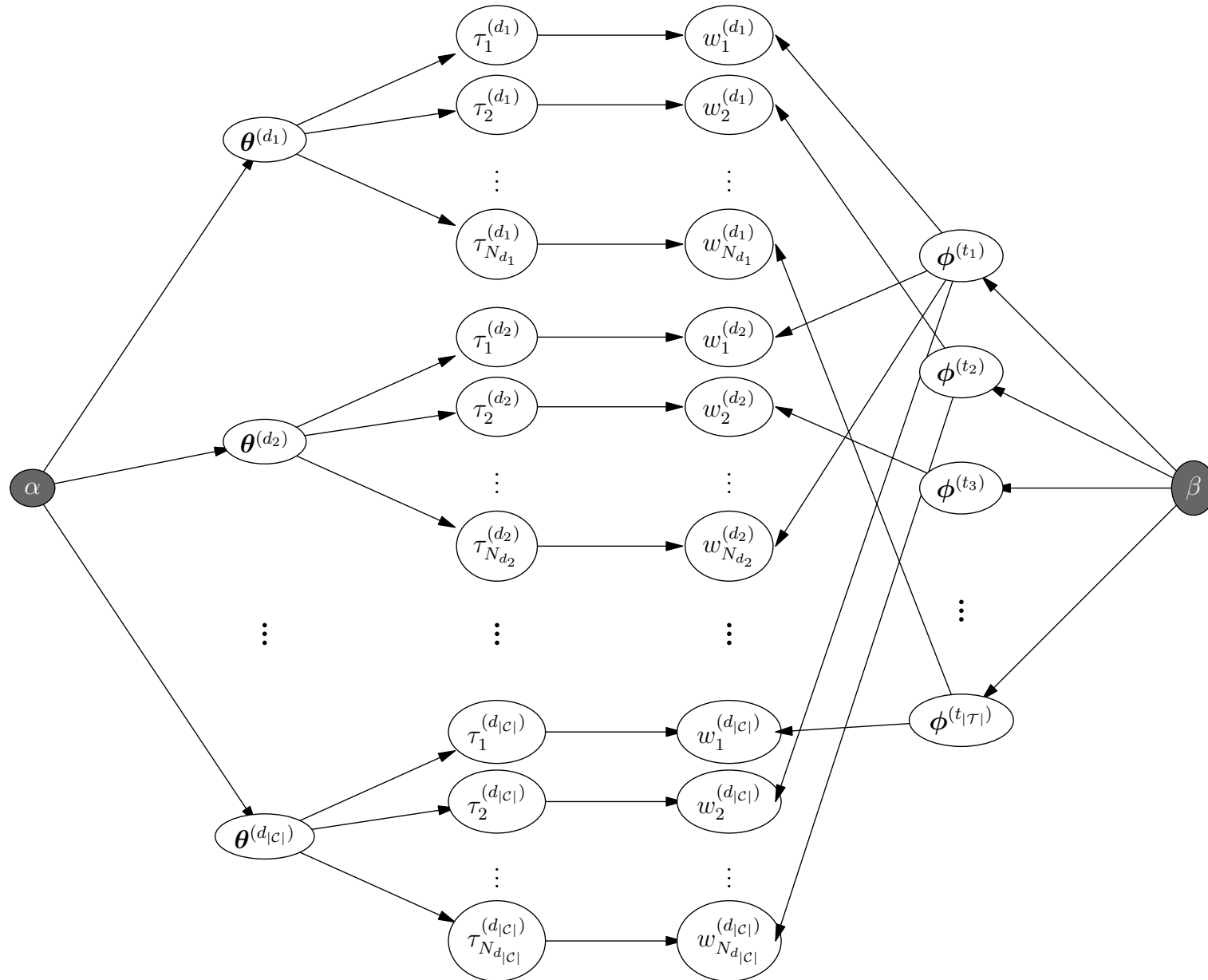
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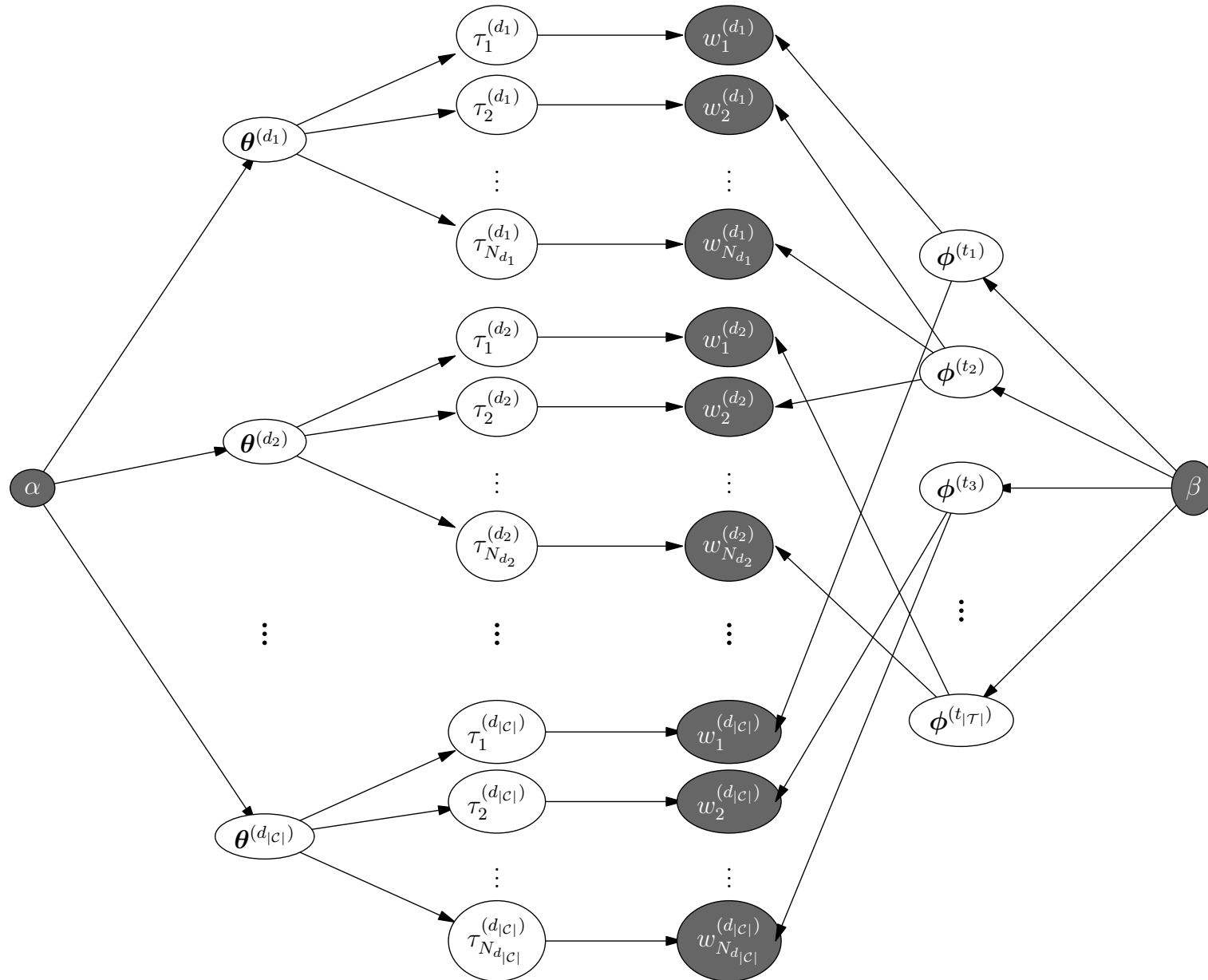
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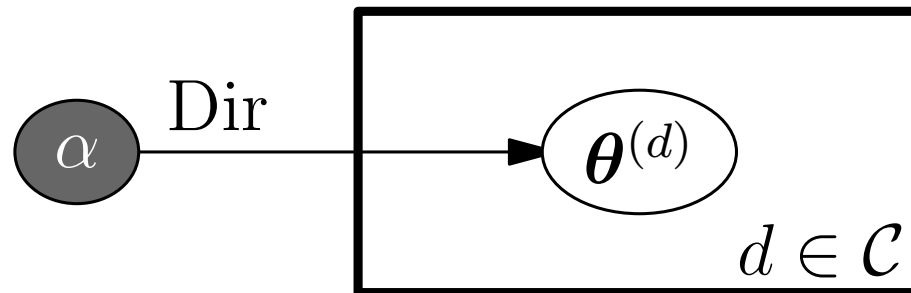
# LDA Graphical Model (version 1)





# Plate Diagrams

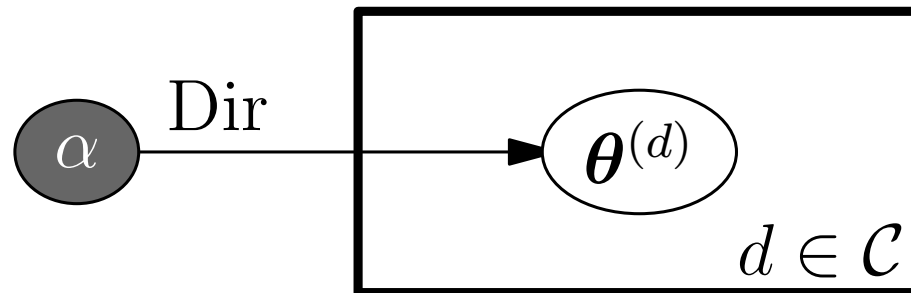
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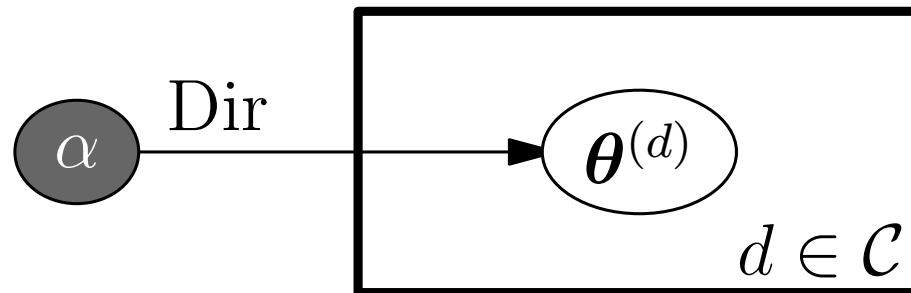
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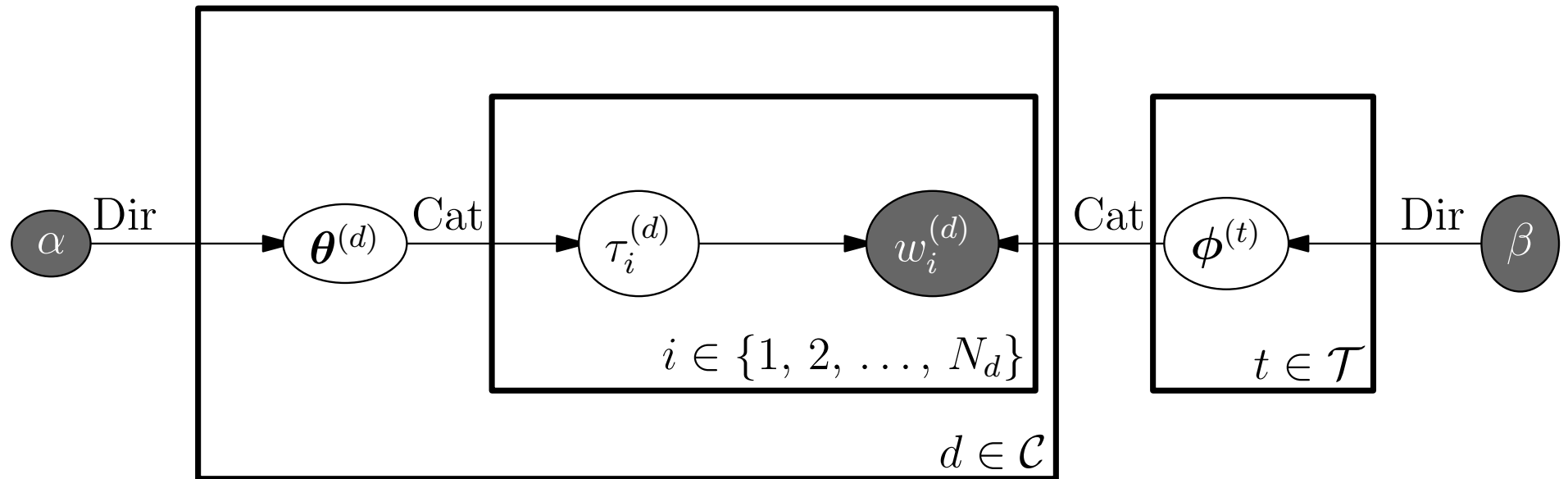
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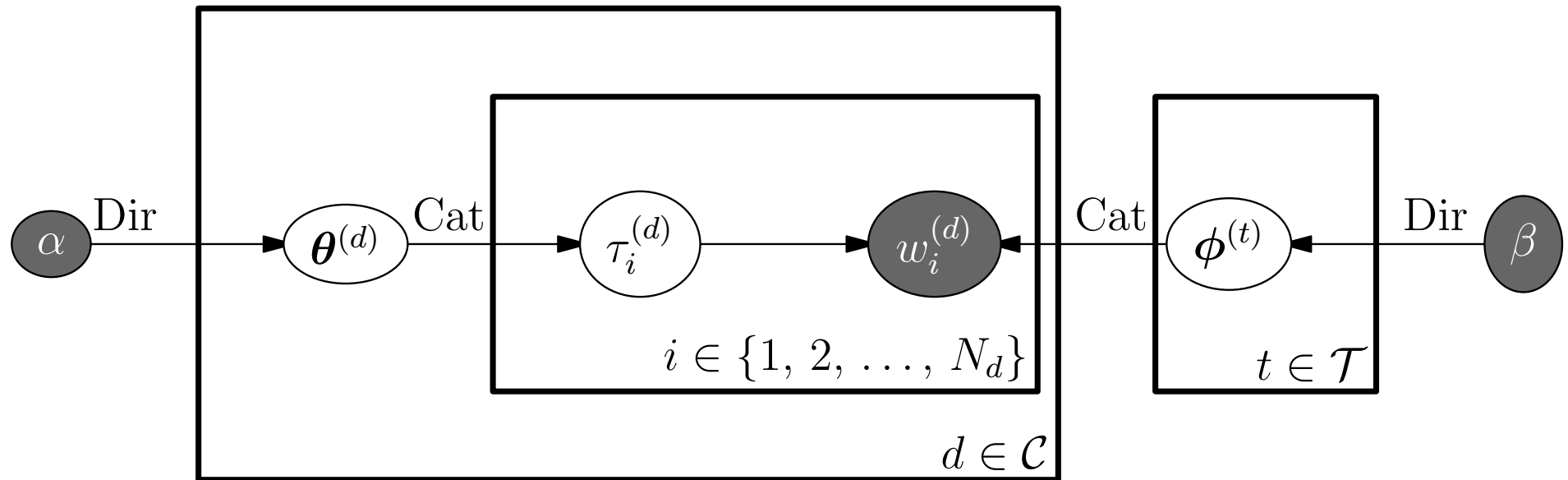
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# Probabilistic Model

- The graphical Model is shorthand for the variables

$$\mathbf{W} = (\mathbf{w}^{(d)} | d \in \mathcal{C}) \quad \text{with} \quad \mathbf{w}^{(d)} = (w_1^{(d)}, w_2^{(d)}, \dots, w_{N_d}^{(d)}), \quad \text{and} \quad w_i^{(d)} \in \mathcal{V}$$

$$\mathbf{T} = (\tau_i^{(d)} | d \in \mathcal{C} \wedge i \in \{1, 2, \dots, N_d\}) \quad \text{with} \quad \tau_i^{(d)} \in \mathcal{T}$$

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- Distributed according to

$$\mathbb{P}(\mathbf{W}, \mathbf{T}, \mathbf{\Theta}, \mathbf{\Phi} | \alpha, \beta) = \left( \prod_{t \in \mathcal{T}} \text{Dir}(\boldsymbol{\phi}^{(t)} | \beta \mathbf{1}) \right) \left( \prod_{d \in \mathcal{C}} \text{Dir}(\boldsymbol{\theta}^{(d)} | \alpha \mathbf{1}) \prod_{i=1}^{N_d} \text{Cat}(\tau_i^{(d)} | \boldsymbol{\theta}^{(d)}) \text{Cat}(w_i^{(d)} | \boldsymbol{\phi}^{(\tau_i^{(d)})}) \right)$$

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# Finding Topics

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# Summary

- Building probabilistic models is an intricate process
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- This allows us to break down the joint probability of all the variables into conditional probabilities
- This is useful for building the model, but also can speed up evaluating expectations
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